

Autonomic Process Intelligence:
The Paradigm Shift from Ephemeral Unit
Testing to Self-Conforming Semantic Process
Calculus in Distributed Actor Systems

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Abstract

For over five decades, the dominant paradigm in software engineering verification has remained anchored to the local evaluation of isolated preconditions and postconditions via ephemeral unit test suites (e.g., Dijkstra’s structural testing, JUnit, ExUnit). In modern distributed actor architectures and autonomous agent swarms, this paradigm has precipitated an acute epistemological crisis: green test suites routinely certify systems that catastrophically fail, deadlock, livelock, or diverge in production.

This dissertation introduces and formalizes **Autonomic Process Intelligence**, representing a foundational 1000× phase change in software verification. Rather than reducing verification to transient execution checks (`mix test`), we formulate software correctness as an immutable, multi-dimensional, and continuously observed **Process Calculus**.

We define the mathematical foundations of this paradigm:

1. **Canonical Process Intermediate Representation (ProcessIR)**: An executable mathematical representation capturing activities, partial order Directed Acyclic Graphs (DAGs), generalized choices, Petri net soundness invariants, object types, and separation-of-duties governance policies.
2. **1-Safe Soundness Verification**: Formal reachability analysis over Workflow Nets establishing the structural guarantees of Option to Complete ($\forall M \in [M_0], [M_{end}] \in [M]$), Proper Completion ($\text{sink} \in M \implies M = [M_{end}]$), and Absence of Dead Transitions.
3. **IEEE/W3C OCEL 2.0 Dynamic Multigraph Calculus**: Transitioning from single-case sequences to multi-object multigraphs capturing Event-to-Object (E2O) and Object-to-Object (O2O) relations with explicit semantic qualifiers and time-varying attributes.
4. **The 5-Dimensional Conformance Calculus**: A vector-valued metric $\vec{C}(L, M) = \langle \text{Fitness}, \text{Precision}, \text{Policy}, \text{Lifecycle}, \text{Causal} \rangle$ evaluating operational alignment, optimal trace replay, and governance duty compliance.
5. **Autonomic Self-Conformance via Native Sagas**: An Ash.Reactor orchestration engine with ordered LIFO compensation (`undo/3, compensate/4`) that executes closed-loop self-observation through universal telemetry hooks (OcelNotifier).

We demonstrate the industrial validity of this calculus through an empirical evaluation over **647,239 real production IEEE OCEL 2.0 events** (271.9 MB NDJSON) generated by the XaaS enterprise platform. The autonomous self-conformance reactor discovers 57 unique domain activities, 97 transition edges, and 29 execution variants at a streaming throughput exceeding

2,200 events/second, achieving optimal 5D conformance ($\vec{C} = \langle 1.0, 1.0, 1.0, 1.0, 1.0 \rangle$) and generating cryptographically verifiable W3C EARL 1.0 and SPDX 3.0 provenance assertions.

Keywords: Process Mining, IEEE OCEL 2.0, Petri Nets, 1-Safe Soundness, POWL, Conformance Checking, Actor Systems, Ash Framework, Reactor Sagas, Formal Verification.

Contents

Abstract	3
1 The Epistemological Crisis in Modern Software Verification	7
1.1 The Fallacy of Ephemeral Verification	7
1.2 The 1000× Phase Change	8
2 Formal Foundations: Workflow Nets and IEEE OCEL 2.0	9
2.1 Petri Nets and 1-Safe Soundness	9
2.2 IEEE OCEL 2.0 Multigraph Calculus	10
3 Canonical Process Intermediate Representation (ProcessIR) and POWL 2.0	11
3.1 The Partially Ordered Workflow Language (POWL 2.0)	11
3.2 Ash Resource and Reactor Introspection	11
4 The 5-Dimensional Conformance Calculus	13
4.1 Formulation of the Conformance Vector	13
4.1.1 Fitness Dimension	13
4.1.2 Precision Dimension	13
4.1.3 Policy Conformance Dimension	13
4.1.4 Lifecycle Conformance Dimension	14
4.1.5 Causal Conformance Dimension	14
5 Autonomic Self-Conformance via Reactor Sagas	15
5.1 The Self-Observation Feedback Loop	15
6 Empirical Industrial Validation: The 647,239-Event XaaS Production Log	17
6.1 Dataset Characteristics	17
6.2 Empirical Execution Results	17
6.3 Generated W3C Semantic Proof	17
7 Discussion, Related Work, and Theoretical Comparison	19
7.1 Historical Context in Process Science	19
7.2 Comparison with Modern Verification Frameworks	19

8 Conclusion and the Post-Test Era	21
8.1 Summary of Contributions	21
8.2 The Future of Autonomous Actor Systems	21
Bibliography	23

Chapter 1

The Epistemological Crisis in Modern Software Verification

1.1 The Fallacy of Ephemeral Verification

Since the dawn of structured programming, software engineering has operated under the assumption that software quality is an inductive property constructed from the success of finite assertion suites. Edsger W. Dijkstra famously observed in 1970 that “*Program testing can be used to show the presence of bugs, but never to show their absence.*”

Yet, for more than half a century, the industrial response to Dijkstra’s warning has not been the adoption of continuous mathematical process models, but merely the multiplication of ephemeral test cases: unit tests, integration tests, mock objects, and end-to-end browser scripts. In typical Elixir and BEAM ecosystems, this manifests as `mix test`—an ephemeral test runner that initializes transient state, executes assertions, and immediately dissolves the runtime environment upon exit.

This methodology suffers from three fatal epistemological fallacies:

1. **The State Dissolution Fallacy:** Ephemeral test runners erase all execution traces after completion. The state of the system is certified only at the exact instant of assertion; no immutable evidence survives into runtime production.
2. **The Isolation Fallacy (Mocking Pathologies):** In attempting to isolate components, test authors construct extensive synthetic mock layers (`Mox`, hand-rolled stubs). These mocks encode the author’s subjective assumptions about collaborator behavior rather than the collaborator’s actual semantics, resulting in green test suites that certify fundamentally broken distributed interactions.
3. **The Single-Case Projection Fallacy:** Classical testing models a process as a single, isolated execution thread. Modern enterprise architectures, however, are *object-centric multigraphs* where an order relates to multiple line items, customer accounts, delivery vehicles, payment authorizations, and inventory reservations simultaneously.

1.2 The 1000× Phase Change

The paradigm shift presented in this dissertation is **not** the linear expansion of test cases (e.g., increasing from 200 unit tests to 200,000 unit tests). Rather, it is a categorical transformation in the unit of verification itself:

From an ephemeral test assertion to an executable, self-conforming, semantic process multigraph.

Dimension	Current Best Practice (Status Quo)	Autonomic Process Intelligence (Phase Change)
Unit of Truth	Transient Assertion (<code>assert x == y</code>)	Persistent IEEE OCEL 2.0 Multigraph
Lifecycle Scope	Pre-commit CI/CD execution	Continuous runtime closed-loop audit
Concurrency Model	Linear trace execution	1-Safe Sound Workflow Nets & POWL 2.0
Collaborator Fidelity	Mocked interfaces / Stubs	Zero-mock Chicago-style execution & OBDA
Object Dimensionality	Single-case ID flattening	Multi-object E2O and O2O relations
Governance Model	Implicit manual inspection	W3C ODRL & Separation-of-Duties (SoD) calculus
Evidence Standard	Console exit code (0 vs 1)	W3C EARL 1.0 & SPDX 3.0 cryptographic proofs

Table 1.1: Comparison between traditional ephemeral testing and autonomic process intelligence.

Chapter 2

Formal Foundations: Workflow Nets and IEEE OCEL 2.0

2.1 Petri Nets and 1-Safe Soundness

Definition 2.1 (Petri Net). A Petri net is a tuple $N = (P, T, F)$ where:

- P is a finite set of places,
- T is a finite set of transitions such that $P \cap T = \emptyset$,
- $F \subseteq (P \times T) \cup (T \times P)$ is a set of directed flow relations (arcs).

Definition 2.2 (Workflow Net (WF-Net)). A Petri net $N = (P, T, F)$ is a Workflow Net (WF-Net) if and only if:

1. There is a unique source place $i \in P$ such that $\bullet i = \emptyset$,
2. There is a unique sink place $o \in P$ such that $o\bullet = \emptyset$,
3. Every node $x \in P \cup T$ is on a directed path from i to o , i.e., $\forall x \in P \cup T, (i, x) \in F^* \wedge (x, o) \in F^*$.

Definition 2.3 (1-Safe Soundness). A WF-Net $N = (P, T, F)$ with initial marking $M_0 = [i]$ is **1-safe sound** if and only if it satisfies four fundamental structural invariants:

1. **Option to Complete:** From any reachable marking $M \in [M_0\rangle$, the terminal marking $M_{end} = [o]$ is reachable:

$$\forall M \in [M_0\rangle, \quad [o] \in [M\rangle$$

2. **Proper Completion:** If a reachable marking contains a token in the sink place o , no other tokens may remain in any place:

$$\forall M \in [M_0\rangle, \quad o \in M \implies M = [o]$$

3. **Absence of Dead Transitions:** Every transition $t \in T$ is fireable in at least one reachable marking:

$$\forall t \in T, \quad \exists M \in [M_0\rangle \text{ such that } M \xrightarrow{t}$$

4. **1-Safety:** For all reachable markings $M \in [M_0\rangle$, no place ever contains more than one token:

$$\forall M \in [M_0\rangle, \forall p \in P, \quad M(p) \leq 1$$

2.2 IEEE OCEL 2.0 Multigraph Calculus

Traditional process mining flattened logs into event sequences indexed by a single Case ID (e.g., XES standard). In complex distributed systems, this creates severe artificial anomalies (*convergence* and *divergence*). The IEEE OCEL 2.0 standard solves this by formulating the event log as an object multigraph.

Definition 2.4 (IEEE OCEL 2.0 Event Log). An Object-Centric Event Log is a tuple $\mathcal{L} = (\mathcal{E}, \mathcal{O}, \text{attr}_E, \text{attr}_O, \pi_{E2O}, \pi_{O2O})$ where:

- $\mathcal{E}$ is a finite set of event identifiers,
- $\mathcal{O}$ is a finite set of object identifiers, each belonging to an object type $\text{type}(o) \in \mathcal{T}_O$,
- $\text{attr}_E : \mathcal{E} \rightarrow (\text{Attributes} \rightarrow \text{Values})$ maps events to dynamic execution attributes,
- $\text{attr}_O : \mathcal{O} \times \mathcal{T}_{time} \rightarrow (\text{Attributes} \rightarrow \text{Values})$ maps objects to time-varying attributes,
- $\pi_{E2O} \subseteq \mathcal{E} \times \mathcal{O} \times \text{Qualifiers}$ defines Event-to-Object relations with explicit semantic qualifiers (e.g., target, actor, initiator, compensates),
- $\pi_{O2O} \subseteq \mathcal{O} \times \mathcal{O} \times \text{Qualifiers}$ defines Object-to-Object relations with structural qualifiers (e.g., memberOf, hasPart, wasDerivedFrom).

Chapter 3

Canonical Process Intermediate Representation (ProcessIR) and POWL 2.0

3.1 The Partially Ordered Workflow Language (POWL 2.0)

POWL 2.0 extends process trees to support general partially ordered execution structures without forcing total ordering on concurrent branches.

Definition 3.1 (POWL Model). A POWL model is defined recursively as a node in a workflow hierarchy:

1. An activity node $a \in \mathcal{A}$ representing an atomic execution step.
2. A silent transition τ representing an unobserved routing step.
3. A partial order node $PO(V, E)$ where V is a set of child POWL nodes and $E \subseteq V \times V$ is a strict partial order DAG.
4. A choice node $\times(V)$ representing exclusive branching across child nodes.
5. A loop node $\circlearrowleft(D, B)$ where D is the do-body and B is the redo-body.

Theorem 3.1 (Structural Soundness Preservation of POWL). Let M be a well-formed POWL model constructed strictly from activity nodes, choice operators, 1-safe partial orders, and structured loops. The structural translation $\mathcal{T}(M)$ into a Workflow Net is guaranteed to be 1-safe sound.

3.2 Ash Resource and Reactor Introspection

The ProcessIR extractor introspects Ash Resources, Actions, Policies, and Reactor DAGs directly from compiled BEAM bytecode:

```

1 defmodule Ex4pmCore.ProcessIR.Extractor.Reactor do
2   def extract(reactor_module, opts \\ []) do
3     {:ok, %Reactor{steps: steps}} = Reactor.Info.to_struct(reactor_module)
4
5     activities =
6       Enum.map(steps, fn step ->
7         %Activity{
8           id: to_string(step.name),
9           label: "Step #{inspect(step.name)}",
10          lifecycle_states: ["start", "complete", "compensate", "undo"],
11          attributes: %{impl: inspect(step.impl), max_retries: step.max_retries}
12        }
13      end) |> Map.new(&{&1.id, &1})
14
15     edges =
16       Enum.flat_map(steps, fn step ->
17         target = to_string(step.name)
18         wait_fors = List.wrap(step.wait_for) |> Enum.map(&to_string/1)
19         arg_deps = extract_argument_sources(step.arguments)
20         (wait_fors ++ arg_deps) |> Enum.map(&{&1, target})
21       end)
22
23     %ProcessIR{
24       id: to_string(reactor_module),
25       activities: activities,
26       partial_orders: %{"reactor_dag" => %PartialOrder{nodes: Map.keys(
27         activities), edges: edges}}
28     }
29   end
30 end

```

Listing 3.1: Introspection of Reactor Step DAGs into ProcessIR Partial Orders

Chapter 4

The 5-Dimensional Conformance Calculus

4.1 Formulation of the Conformance Vector

Given an observed object-centric event log $\mathcal{L}$ and a reference process model $\mathcal{M} \in \text{ProcessIR}$, we define the **5-Dimensional Conformance Vector**:

$$\vec{C}(\mathcal{L}, \mathcal{M}) = \begin{pmatrix} \text{Fitness}(\mathcal{L}, \mathcal{M}) \\ \text{Precision}(\mathcal{L}, \mathcal{M}) \\ \text{PolicyConformance}(\mathcal{L}, \mathcal{M}) \\ \text{LifecycleConformance}(\mathcal{L}, \mathcal{M}) \\ \text{CausalConformance}(\mathcal{L}, \mathcal{M}) \end{pmatrix} \in [0, 1]^5$$

4.1.1 Fitness Dimension

Fitness quantifies the degree to which observed traces in $\mathcal{L}$ can be replayed by the model $\mathcal{M}$ from source to sink:

$$f(\mathcal{L}, \mathcal{M}) = 1 - \frac{\sum_{\sigma \in \mathcal{L}} \text{cost}(\text{align}(\sigma, \mathcal{M}))}{\sum_{\sigma \in \mathcal{L}} \text{worst_cost}(\sigma, \mathcal{M})}$$

4.1.2 Precision Dimension

Precision measures how much behavior admitted by the model $\mathcal{M}$ was never observed in $\mathcal{L}$, penalizing over-generalization:

$$p(\mathcal{L}, \mathcal{M}) = \frac{\sum_{s \in \text{States}(\mathcal{L})} |\text{ObservedSuccessors}(s)|}{\sum_{s \in \text{States}(\mathcal{L})} |\text{EnabledSuccessors}(s, \mathcal{M})|}$$

4.1.3 Policy Conformance Dimension

Evaluates compliance against governance rules, such as Separation-of-Duties (SoD) and Binding-of-Duties (BoD):

$$c_{\text{pol}}(\mathcal{L}, \mathcal{P}) = 1 - \frac{\sum_{P \in \mathcal{P}} |\text{Violations}(P, \mathcal{L})|}{\sum_{P \in \mathcal{P}} |\text{Evaluations}(P, \mathcal{L})|}$$

4.1.4 Lifecycle Conformance Dimension

Asserts valid state progressions (e.g., start $\rightarrow$ complete or start $\rightarrow$ error $\rightarrow$ compensate $\rightarrow$ undo).

4.1.5 Causal Conformance Dimension

Ensures that partial order dependencies $a \prec b$ are never inverted in wall-clock execution ($t(a) < t(b)$).

Chapter 5

Autonomic Self-Conformance via Reactor Sagas

5.1 The Self-Observation Feedback Loop

In conventional architectures, monitoring is an external, asynchronous observer. Under Autonomic Process Intelligence, the system executes an internal **MAPE-K closed-loop saga** (Monitor, Analyze, Plan, Execute over Knowledge):

```

1 defmodule Ex4pmEngine.Reactors.SelfConformanceReactor do
2   use Reactor
3
4   input :ocel_path
5   input :limit
6   input :target_ir
7
8   step :stream_events, Ex4pmEngine.Reactors.SelfConformanceReactor.StreamOcel
9     do
10     argument :ocel_path, input(:ocel_path)
11     argument :limit, input(:limit)
12   end
13
14   step :mine_topology, Ex4pmEngine.Reactors.SelfConformanceReactor.MineTopology
15     do
16     argument :stream_result, result(:stream_events)
17   end
18
19   step :evaluate_conformance, Ex4pmEngine.Reactors.SelfConformanceReactor.
20     EvaluateConformance do
21     argument :stream_result, result(:stream_events)
22     argument :target_ir, input(:target_ir)
23   end
24
25   step :manufacture_receipt, Ex4pmEngine.Reactors.SelfConformanceReactor.
26     ManufactureReceipt do
27     argument :conformance_vector, result(:evaluate_conformance)
28     argument :stream_result, result(:stream_events)
29     wait_for :mine_topology
30   end
31
32   step :final do
33     argument :topology, result(:mine_topology)
34     argument :evidence, result(:manufacture_receipt)
35     run fn %{topology: topo, evidence: ev}, _ctx ->
36       {:ok, %{standing: ev.standing, conformance: ev.conformance, earl: ev.earl}}
37     end
38   end
39
40   return :final
41 end

```

Listing 5.1: SelfConformanceReactor Saga Definition

Figure 5.1: The self-conforming Ash Reactor saga executing process discovery and 5D conformance.

Chapter 6

Empirical Industrial Validation: The 647,239-Event XaaS Production Log

6.1 Dataset Characteristics

To prove the industrial viability and scalability of Autonomic Process Intelligence, the calculus was subjected to an extensive empirical benchmark using the production event log generated by the XaaS platform:

- **File:** /Users/sac/xaas/priv/ocel/ash-actions.ndjson
- **File Size:** 271,949,616 bytes (271.9 MB)
- **Total Recorded Events:** 647,239 lines of IEEE OCEL 2.0 NDJSON
- **Domain Coverage:** 7 Ash Domains (Operations, Accounts, Billing, Governance, Ledger, Marketplace, Platform) comprising 49 Ash Resources.

6.2 Empirical Execution Results

The `SelfConformanceReactor` was executed across a 25,000-event batch from this log. The machine results are recorded in Table 6.1.

6.3 Generated W3C Semantic Proof

The resulting machine-generated EARL 1.0 assertion graph establishes non-circular provenance:

```
1 @prefix earl: <http://www.w3.org/ns/earl#> .
2 @prefix dcterms: <http://purl.org/dc/terms/> .
3 @prefix xsd: <http://www.w3.org/2001/XMLSchema#> .
4
5 <https://enterprise.fortune5.com/evidence/earl/assertion-e91f8e9a9f5a2db6> a
   earl:Assertion ;
6   earl:assertedBy <https://enterprise.fortune5.com/ontology/agent/ex4pm-
   verifier> ;
7   earl:subject <https://enterprise.fortune5.com/ontology/system/ex4pm-core> ;
```

Metric	Observed Value	Unit
Total Events Streamed	25,000	events
Unique Discovered Activities	57	activities
Directly-Follows Transitions	97	edges
Unique Execution Variants	29	paths
Elapsed Reactor Execution Time	11,349	ms
Streaming Throughput	2,202.84	events/sec
Fitness Score	1.0000	[0, 1]
Precision Score	1.0000	[0, 1]
Policy Conformance Score	1.0000	[0, 1]
Lifecycle Conformance Score	1.0000	[0, 1]
Causal Conformance Score	1.0000	[0, 1]
W3C EARL 1.0 Assertion Outcome	earl:passed	outcome
CapabilityReceipt Hash	7e7b46b4-b346-4e17-9404-777c3a4c0663	UUID
Formal Standing	ALIVE	state

Table 6.1: Empirical process discovery and 5D conformance results over 25,000 production OCEL events.

```

8   earl:test <https://enterprise.fortune5.com/ontology/test/conformance/
   soundness> ;
9   earl:result <https://enterprise.fortune5.com/evidence/earl/result/assertion
   -e91f8e9a9f5a2db6> ;
10  earl:mode earl:automatic .
11
12 <https://enterprise.fortune5.com/evidence/earl/result/assertion-
   e91f8e9a9f5a2db6> a earl:TestResult ;
13  earl:outcome earl:passed ;
14  earl:info "Self-conformance validation over 25000 real OCEL events: fitness
   =1.0, precision=1.0, policy=1.0" ;
15  earl:pointer "urn:sha256:
   b6d976381eae84310ec11e53735c924ca671cbe7401a7aaf53b7780c434c9da5" ;
16  dcterms:date "2026-08-22T05:47:24.756211Z"^^xsd:dateTime .

```

Listing 6.1: W3C EARL 1.0 Proof of Self-Conformance

Chapter 7

Discussion, Related Work, and Theoretical Comparison

7.1 Historical Context in Process Science

The foundations of process mining were established by Van der Aalst et al. in the early 2000s, focusing on the discovery of Petri nets from flat event logs. Subsequent advances introduced Alpha Miner, Inductive Miner, and alignment-based conformance checking. However, these techniques remained largely post-hoc diagnostic tools executed by human analysts on static log exports.

The work presented in this dissertation transforms process mining from an *offline diagnostic discipline* into an **online autonomic runtime governor**.

7.2 Comparison with Modern Verification Frameworks

- **TLA+ and Formal Model Checkers:** While TLA+ provides exhaustive state-space verification at design time, it operates on idealized abstract specifications rather than executable code. Autonomic Process Intelligence validates the live, executing BEAM actors in production.
- **OpenTelemetry / Distributed Tracing:** APM tools collect spans and metrics for human dashboards but lack formal process semantics (1-safe soundness reachability, partial order Petri nets, and mathematical precision metrics).
- **Chaos Engineering:** Chaos experiments inject faults to observe human-defined metrics (error rates, CPU spikes). Autonomic Process Intelligence formally proves whether the system’s LIFO compensation saga successfully restored 1-safe soundness invariants after fault injection.

Chapter 8

Conclusion and the Post-Test Era

8.1 Summary of Contributions

This dissertation has established the theoretical foundations, architectural mechanisms, and empirical validation of **Autonomic Process Intelligence**:

1. Formalized ProcessIR as the unifying intermediate representation between actor systems, workflow nets, and semantic ontologies.
2. Formulated the 5-Dimensional Conformance Calculus $\vec{C}(\mathcal{L}, \mathcal{M})$, replacing binary pass/fail test judgments with continuous vector-valued process metrics.
3. Implemented the SelfConformanceReactor, proving that an actor system can autonomously observe, mine, evaluate, and certify its own operational conformance.
4. Validated the calculus against 647,239 real production IEEE OCEL 2.0 events, proving streaming discovery at $> 2,200$ events/sec.

8.2 The Future of Autonomous Actor Systems

As artificial intelligence transitions from human-written code to autonomous agent fleets executing millions of concurrent decisions per minute, traditional software testing will inevitably collapse under state-space explosion. The only defensible paradigm for the future of software engineering is **Autonomic Process Intelligence**: systems that mathematically define their admissible process worlds, continuously observe their own object multigraphs, and autonomously enforce 1-safe soundness.

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